

Nonmaximal sums of maximally monotone operators under Rockafellar's constraint qualification

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Abstract

We construct counterexamples to Rockafellar's sum conjecture in which two maximally monotone operators satisfy the interior-domain condition but their sum is not maximally monotone. The examples are defined on c_0 and standard ℓ^1 , with one summand a bounded positive rank-one operator defined on the whole space. We first compute the entire monotone polar of a class of graphs and characterize their maximal monotonicity by the solvability of explicit equations in the original continuous dual. For the construction on c_0 , we couple triangular operators through a Lipschitz curve of block sums. The positive-index coordinates of each curve value have finite support, and their common limit at the two deleted endpoints belongs to $\ell^2 \setminus \ell^1$. The polar formula shows that any additional polar point would require these nonsummable coordinates in the continuous dual, proving maximality. An exact pairing identity then proves nonmaximality of the sum and a finite radial bound for the first operator. A bounded surjection transfers the construction to standard ℓ^1 , where maximality is proved in the full $\ell^1 \times \ell^\infty$ pair.

Keywords: maximally monotone operator; sum theorem; nonreflexive Banach space; monotone polar; rank-one operator.

1 Introduction

Let X be a real Banach space, let X^* be its continuous dual, and let $A, B : X \rightrightarrows X^*$ be maximally monotone operators. Their pointwise sum is defined by $(A + B)x = \{a + b : a \in Ax, b \in Bx\}$. Rockafellar's sum theorem [1] states that $A + B$ is maximally monotone when X is reflexive and

$$\text{dom } A \cap \text{int}(\text{dom } B) \neq \emptyset. \quad (1)$$

The unrestricted sum question asks whether Eq. (1) alone is sufficient on an arbitrary real Banach space. The interior here is taken in the norm topology of X , and maximality is taken in the full pair $X \times X^*$. The broader theory of maximality and convex representation on Banach spaces is developed in Simons' monograph [2].

Several positive results extend the theorem by imposing structure on one of the factors. Yao [3] proves maximality when the first factor is of type (FPV) and the second factor has full domain. Borwein and Yao [4] prove the sum theorem under Eq. (1) when the first factor is a linear relation. The factor order in the latter qualification matters: placing a full-domain linear operator first would require interior points in the domain of the other factor. These results motivate the question of whether an elementary full-domain second factor can already produce a nonmaximal sum with a nonlinear first factor. Normal-cone partners under type (FPV) hypotheses are treated by Voisei [5]. At the universal level, Boţ, Bueno, and Simons [6] show that restricting the sum question to pairs with bounded common domain does not weaken it. This is a statement about all admissible pairs, not a localization theorem for one fixed factor.

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Convex representations provide a second way to organize these questions. Fitzpatrick [7] associates a convex function with a monotone graph. Burachik and Svaiter [8] study the associated family of convex representatives through enlargements, and Penot [9] develops representation methods for compositions and sums in the reflexive setting. For nonmaximal graphs, representability and maximality must still be distinguished; Martínez-Legaz and Svaiter [10] study this distinction, including unique maximal extensions in finite dimensions. Bueno, Martínez-Legaz, and Svaiter [11] compare representable and monotone-polar closures. Our structural theorem concerns graphs containing every point in specified affine fibres and computes their polar directly; it does not replace these representation theories by an unrestricted normal form. Marques Alves and Svaiter [12] prove a sum theorem under a qualification on domains of Fitzpatrick representations. Such a qualification must be checked on those representations; it cannot be replaced without proof by Eq. (1).

Nonreflexive constructions provide a concrete setting for the sum question. Gossez's work [13] develops monotone-operator theory beyond reflexive spaces. The dense-type class has additional structure: Marques Alves and Svaiter [14] identify types (D) and (NI), and Bauschke, Borwein, Wang, and Yao [15] prove that Fitzpatrick–Phelps type (FP) implies dense type. Type (FP) here is not type (FPV). These results distinguish important subclasses; none of these type assumptions is imposed in the question above. Bueno and Svaiter [16] construct a non-type (D) maximally monotone operator on c_0 . Bauschke, Borwein, Wang, and Yao [17] develop a family of triangular operators and further nonlinear examples. In particular, Example 4.1 of their August 6, 2011 author version contains the single-block triangular map used below. These earlier ingredients give exact pairing identities. To obtain the sum behavior considered here, we impose an additional constraint on the sums over countably many blocks and verify maximality for every point of the full monotone polar.

We define this constraint through a 1-Lipschitz map $F : J \rightarrow \ell^1(\mathbb{N}_0) \subset \ell^2(\mathbb{N}_0)$, where $J = (-\infty, -1) \cup (1, \infty)$. Its positive-index coordinates have finite support for each actual parameter. At both deleted endpoints they tend, coordinatewise and in ℓ^2 , to $(1/(4n))_{n \geq 1}$, which is not in ℓ^1 . The triangular pairing identity first proves monotonicity. The complete affine fibres of the graph then reduce the condition for an arbitrary polar point to

$$\|F(t) - m_p\|_2^2 + \nu^2 \leq (t - \tau)^2 \quad (t \in J), \quad (2)$$

where p is the dual component of the polar point, $m_p = (\sum_j p_{b,j})_{b \geq 0} \in \ell^1(\mathbb{N}_0)$ is its sequence of block sums, and $\tau, \nu \in \mathbb{R}$. If $|\tau| > 1$, the test $t = \tau$ returns a graph point. If $|\tau| \leq 1$, limits in ℓ^2 at the two endpoints force the positive-index coordinates of m_p to equal $1/(4n)$. Their nonsummability excludes this second case and proves maximality.

The resulting first factor A has nonempty domain at positive distance from zero. Its construction also gives $\langle x, a \rangle > -\sigma$ on $\text{gra } A$, with $\sigma = \pi^2/96 < 1/8$. We choose $Px = \langle x, g \rangle g$ for a specified nonzero $g \in (c_0)^*$. This factor is also the subdifferential of the continuous convex quadratic $x \mapsto \langle x, g \rangle^2/2$, consistent with Rockafellar's subdifferential maximality theorem [18]. We give a direct maximality proof for P below. The same pairing identity gives $\langle x, a + Px \rangle > 1 - \sigma$ for every $(x, a) \in \text{gra } A$. Thus $(0, 0)$ is monotonically related to the entire sum and is not in its graph. The full domain of P makes Eq. (1) immediate from nonemptiness of $\text{dom } A$.

1.1 Contributions

The contributions of this paper are as follows.

(1) We give a counterexample to Rockafellar's sum conjecture on c_0 (Example 1). The second operator is bounded, positive, rank-one and defined everywhere. The first operator has nonempty domain, so the original interior-domain condition holds. We prove maximality of both operators and exhibit a point monotonically related to their sum that does not belong to its graph.

(2) We establish a formula for the entire monotone polar of the graph class in Eq. (8) and a necessary and sufficient condition for maximality (Theorem 1). The proof separates the roles of the pairing identity, the complete affine fibres and the original continuous dual. It also yields a finite radial bound and the rank-one perturbation that makes the sum nonmaximal. We verify all hypotheses in the coordinate construction and derive nonconvexity of the first operator's domain closure (Corollary 1).

(3) We construct a counterexample on standard ℓ^1 (Example 2) by pulling back the first operator through a specified bounded surjection onto c_0 . Tests along the kernel force every polar dual component to factor through the surjection, proving maximality against the entire ℓ^∞ dual. This transfers the same failure mechanism to a second Banach space.

The paper is organized as follows. Section 2 introduces the notation and basic definitions. Section 3 proves the structural theorem and the pullback lemma. Section 4 constructs the counterexamples on c_0 and standard ℓ^1 and derives their stated consequences. Section 5 concludes the paper.

2 Preliminaries

All scalars are real. Put $\mathbb{N} = \{1, 2, \dots\}$ and $\mathbb{N}_0 = \{0, 1, \dots\}$. The continuous dual of a Banach space X is denoted by X^* , and $\langle x, a \rangle$ denotes its duality pairing. We use the norm topology for interiors and closures unless stated otherwise.

Definition 1 (Monotonicity and the monotone polar). For a graph $G \subset X \times X^*$, define

$$G^\mu = \{(z, p) \in X \times X^* : \langle z - x, p - a \rangle \geq 0 \quad \forall (x, a) \in G\}. \quad (3)$$

The graph is monotone when $\langle x - y, a - b \rangle \geq 0$ for all $(x, a), (y, b) \in G$, and maximally monotone when it has no proper monotone extension in this same full dual pair.

For a monotone G , $G = G^\mu$ is equivalent to maximality: a point in $G^\mu \setminus G$ can be adjoined to G , and every point of a monotone extension belongs to G^μ . We write $\text{dom } G = \{x : \exists a, (x, a) \in G\}$ and $\text{ran } G = \{a : \exists x, (x, a) \in G\}$.

Definition 2 (Radial bound at the origin). For a nonempty graph G with $0 \notin \text{dom } G$, define

$$\mathcal{V}(G) = \sup_{(x, a) \in G} \frac{\max\{0, -\langle x, a \rangle\}}{\|x\|}. \quad (4)$$

At $(0, 0)$ this is the nonnegative slope used by Verona and Verona [19]. The finite upper bounds proved below are properties of the constructed graphs, not an assumed general bound from that source.

3 Maximal monotonicity and nonmaximal sums

We first isolate the properties that determine the entire monotone polar. The bilinear identity below makes a Lipschitz constraint monotone, and the complete affine fibres reduce arbitrary polar points to a scalar comparison. The two endpoint comparisons then identify exactly which additional continuous-dual labels could extend the graph. The theorem concerns the stated class of graphs; its assumptions will be verified for the coordinate construction.

Let X be a real Banach space and let H_0 be a real Hilbert space. In this section put $H = \mathbb{R} \oplus H_0$. Let $E : X^* \rightarrow X$ and $M : X^* \rightarrow H$ be bounded linear maps, and let $g \in X^* \setminus \{0\}$ satisfy $Mg = 0$ and

$$\langle Ea, b \rangle + \langle Eb, a \rangle = 2\langle Ma, Mb \rangle_H \quad (a, b \in X^*). \quad (5)$$

Define $h = Eg$, $r(a) = \langle h, a \rangle$, $La = -Ea + r(a)h$, and $K = \ker M \cap \ker r$. Assume the exact primal annihilator identity

$$\{x \in X : \langle x, k \rangle = 0 \text{ for all } k \in K\} = \mathbb{R}h. \quad (6)$$

Let $\omega : (0, \infty) \rightarrow H_0$ be ℓ -Lipschitz, where $0 \leq \ell \leq 1$, and suppose

$$\omega(s) \longrightarrow \eta \text{ in } H_0 \quad (s \downarrow 0), \quad 0 < S := \|\eta\|^2 < 1, \quad \|\omega(s)\|^2 \leq S. \quad (7)$$

Put $J = (-\infty, -1) \cup (1, \infty)$ and $F(t) = (\operatorname{sgn} t, \omega(|t| - 1))$. For every $t \in J$, assume there exists an actual $a^t \in X^*$ with $r(a^t) = t$ and $Ma^t = F(t)$. We include every $a \in X^*$ with these values, defining

$$D = \{a \in X^* : r(a) \in J, Ma = F(r(a))\}, \quad G = \{(La, a) : a \in D\}. \quad (8)$$

In particular, each parameter fibre is $a^t + K$, without a boundedness or continuity assumption on the choice $t \mapsto a^t$.

Theorem 1 (Monotone polar and maximality). *Under the preceding assumptions, define*

$$\widehat{F}(t) = \begin{cases} F(t), & |t| > 1, \\ (t, \eta), & |t| \leq 1. \end{cases}$$

Then the entire monotone polar in $X \times X^$ is*

$$G^\mu = \{(Lp, p) : p \in X^*, Mp = \widehat{F}(r(p))\}. \quad (9)$$

It is the unique maximally monotone extension of G . Moreover, G itself is maximally monotone if and only if

$$\nexists (p, \tau) \in X^* \times [-1, 1] : \quad r(p) = \tau, \quad Mp = (\tau, \eta). \quad (10)$$

When Eq. (10) holds, the operator $\mathcal{A}$ with graph G satisfies $0 \notin \operatorname{dom} \mathcal{A}$ and $\mathcal{V}(G) \leq S\|g\|$. With $\mathcal{P}x = \langle x, g \rangle g$, both $\mathcal{A}$ and $\mathcal{P}$ are maximally monotone, satisfy $\operatorname{dom} \mathcal{A} \cap \operatorname{int} \operatorname{dom} \mathcal{P} \neq \emptyset$, and have a nonmaximal sum: $(0, 0)$ belongs to its monotone polar and lies outside its graph.

Proof. Equation (5) at $a = b = g$ gives $r(g) = 0$. Using $Mg = 0$ again gives

$$\langle Ea, g \rangle = -r(a), \quad \langle La, g \rangle = r(a), \quad \langle La - Lb, a - b \rangle = |r(a) - r(b)|^2 - \|Ma - Mb\|^2. \quad (11)$$

The equality $r(a^2) = 2$ also proves $h \neq 0$.

Extend ω to zero by $\omega(0) = \eta$; its Lipschitz bound persists by the norm limit. Define $\chi(t) = \max\{-1, \min\{1, t\}\}$ and $\zeta(t) = \max\{|t| - 1, 0\}$. Then $\widehat{F}(t) = (\chi(t), \omega(\zeta(t)))$. If t, u lie on the same exterior branch, in the middle interval, or one in the middle and one outside, the nonnegative changes in χ and ζ add to $|t - u|$. On opposite exterior branches they sum to $2 + ||t| - |u|| \leq |t - u|$. Thus in all cases

$$|\chi(t) - \chi(u)| + |\zeta(t) - \zeta(u)| \leq |t - u|,$$

and

$$\|\widehat{F}(t) - \widehat{F}(u)\|^2 \leq |\chi(t) - \chi(u)|^2 + \ell^2 |\zeta(t) - \zeta(u)|^2 \leq |t - u|^2.$$

Equation (11) proves monotonicity of both G and the set on the right of Eq. (9). It also proves that this latter set is contained in G^μ .

For the reverse inclusion, take an arbitrary $(x, p) \in G^\mu$, where p belongs to the full continuous dual. For $a \in D$ and $t = r(a)$, expansion using Eq. (5) gives

$$\langle x - La, p - a \rangle = \langle x, p \rangle - \langle x + Ep, a \rangle + 2(Ma, Mp) - tr(p) + t^2 - \|Ma\|^2.$$

Replacing a by $a + \theta k$, for all $\theta \in \mathbb{R}$ and $k \in K$, forces $\langle x + Ep, k \rangle = 0$. Hence $x = -Ep + vh$ by Eq. (6). With

$$\tau = \frac{v + r(p)}{2}, \quad \nu = \frac{v - r(p)}{2},$$

the same expansion reduces exactly to

$$\langle x - La, p - a \rangle = (t - \tau)^2 - \nu^2 - \|F(t) - Mp\|^2. \quad (12)$$

Every $t \in J$ is an actual test. If $|\tau| > 1$, testing $t = \tau$ forces $\nu = 0$ and $Mp = F(\tau)$; thus $v = r(p) = \tau$ and $x = Lp$.

If $|\tau| \leq 1$, write $Mp = (b, z) \in \mathbb{R} \oplus H_0$. Limits along the two actual branches yield

$$(b - 1)^2 + \|z - \eta\|^2 + \nu^2 \leq (1 - \tau)^2, \quad (b + 1)^2 + \|z - \eta\|^2 + \nu^2 \leq (1 + \tau)^2.$$

Multiply these by $(1 + \tau)/2$ and $(1 - \tau)/2$, respectively, and add. Expansion gives

$$(b - \tau)^2 + \|z - \eta\|^2 + \nu^2 \leq 0.$$

The endpoint weights remain valid at $\tau = \pm 1$. Consequently $Mp = (\tau, \eta)$, $r(p) = v = \tau$, and $x = Lp$. This proves Eq. (9). The norm limits were taken in H , not in X^* ; the label p was actual from the start.

We have proved that G^μ is monotone. Any point related to it is related to G and therefore belongs to G^μ , so G^μ is maximally monotone. Every monotone extension of G lies in G^μ ; hence a maximal one must equal it. The missing part $G^\mu \setminus G$ consists exactly of the points (Lp, p) for which $r(p) \in [-1, 1]$ and $Mp = (r(p), \eta)$, as in Eq. (10). This proves the criterion.

Finally, each $a \in D$ gives

$$|\langle La, g \rangle| = |r(a)| > 1, \quad \|La\| > 1/\|g\|, \quad \langle La, a \rangle = r(a)^2 - 1 - \|\omega(|r(a)| - 1)\|^2 > -S.$$

Thus $0 \notin \text{dom } \mathcal{A}$ and, for every $a \in D$,

$$\frac{\max\{0, -\langle La, a \rangle\}}{\|La\|} \leq \frac{S}{\|La\|} < S\|g\|.$$

Taking the supremum proves the stated whole-graph radial bound. The map $\mathcal{P}$ is bounded, linear, positive, nonzero, and full-domain. Its monotonicity follows from $\langle x - y, \mathcal{P}x - \mathcal{P}y \rangle = \langle x - y, g \rangle^2$. For an arbitrary point (z, z^*) in its polar, tests at $z + tv$ give

$$-t\langle v, z^* - \mathcal{P}z \rangle + t^2\langle v, g \rangle^2 \geq 0 \quad (t \in \mathbb{R}, v \in X).$$

Both signs of arbitrarily small t force $z^* = \mathcal{P}z$. Thus $\mathcal{P}$ is maximally monotone, and $La^2 \in \text{dom } \mathcal{A}$ proves $\text{dom } \mathcal{A} \cap \text{int dom } \mathcal{P} \neq \emptyset$. For every $a \in D$,

$$\langle La, a + \mathcal{P}La \rangle = 2r(a)^2 - 1 - \|\omega(|r(a)| - 1)\|^2 > 1 - S > 0.$$

The sum is monotone and its domain does not contain zero. Adjoining $(0, 0)$ is a proper monotone extension. $\square$

The three conclusions use different parts of the argument. The Lipschitz bound and Eq. (11) prove monotonicity. The complete fibres and Eq. (10) prove maximality in the original dual pair. Finally, the rank-one factor adds $r(a)^2$ to the pairing, giving a positive bracket with the missing origin at every point of the sum graph. A limit of profiles in H contributes a polar point only when an actual element of X^* realizes both of its prescribed coordinates.

The following pullback argument is used only to transport the construction. We include its proof to specify the role of arbitrary continuous-dual labels and of the controlled preimages.

Lemma 1 (Pullback by a bounded surjection). *Let U, V be real Banach spaces and let $Q : U \rightarrow V$ be a bounded linear surjection. Suppose $C_Q > 0$ and every $y \in V$ has a preimage $u \in U$ with $\|u\| \leq C_Q\|y\|$. If $\mathcal{M} : V \rightrightarrows V^*$ is maximally monotone, then*

$$\text{gra}(Q^*\mathcal{M}Q) = \{(u, Q^*a) : (Qu, a) \in \text{gra } \mathcal{M}\}$$

is maximally monotone in the full pair $U \times U^$.*

Proof. A maximally monotone graph is nonempty, since the empty graph admits a one-point monotone extension. Surjectivity therefore makes the pullback graph nonempty. For two of its points,

$$\langle u - v, Q^*(a - b) \rangle = \langle Qu - Qv, a - b \rangle \geq 0,$$

so it is monotone.

Let $(v, v^*) \in U \times U^*$ be related to this entire graph. Fix (u, Q^*a) in the graph. For each $k \in \ker Q$, all $(u + tk, Q^*a)$ belong to it, and their polar inequalities have the form

$$\langle v - u, v^* - Q^*a \rangle - t\langle k, v^* \rangle \geq 0 \quad (t \in \mathbb{R}).$$

Thus v^* annihilates $\ker Q$. Define $p(y) = \langle u, v^* \rangle$ when $Qu = y$. Differences of preimages prove well-definedness, and linear combinations of preimages prove linearity. The prescribed lifting bound gives $|p(y)| \leq C_Q \|v^*\| \|y\|$, so $p \in V^*$ and $v^* = Q^*p$. For every $(y, a) \in \text{gra } \mathcal{M}$, choose $u \in U$ with $Qu = y$. Then (u, Q^*a) belongs to the pullback graph. The remaining inequalities are

$$\langle Qv - y, p - a \rangle \geq 0 \quad ((y, a) \in \text{gra } \mathcal{M}).$$

Maximality of $\mathcal{M}$ gives $(Qv, p) \in \text{gra } \mathcal{M}$, and hence (v, v^*) belongs to the pullback graph. $\square$

We next use Theorem 1 to construct a counterexample on c_0 and then apply Lemma 1 to obtain one on standard ℓ^1 .

4 Counterexamples on c_0 and standard ℓ^1

We first verify the structural theorem for explicit operators on c_0 . A bounded surjection then defines operators on standard ℓ^1 , whose maximality and sum failure are proved in the full continuous dual pair.

4.1 A counterexample on c_0

Put $I = \mathbb{N}_0 \times \mathbb{N}$ and $Y = c_0(I)$ with the supremum norm. For every $x \in Y$ and $\epsilon > 0$, only finitely many $(b, j) \in I$ satisfy $|x_{b,j}| \geq \epsilon$. Its full continuous dual is $\ell^1(I)$, with $\langle x, a \rangle = \sum_{b,j} x_{b,j} a_{b,j}$. We also use this pairing for $x \in \ell^\infty(I)$ and $a \in \ell^1(I)$. The unit coordinate vectors are denoted by $e_{b,j}$.

We index coordinates by blocks. The map m records the sum in each block, and E applies a triangular operator within that block. Its single-block formula is the $\alpha = (1, 1, \dots)$ case of [17, author version, Example 4.1]; the additional constraint below couples the block sums through F .

Definition 3 (Block maps). For $a \in \ell^1(I)$, define

$$(ma)_b = \sum_{j \geq 1} a_{b,j}, \quad (Ea)_{b,j} = a_{b,j} + 2 \sum_{k > j} a_{b,k}. \quad (13)$$

Here $m : \ell^1(I) \rightarrow \ell^1(\mathbb{N}_0) \subset \ell^2(\mathbb{N}_0)$ and $E : \ell^1(I) \rightarrow Y$. Put

$$\begin{aligned} g &= e_{0,1} - e_{0,2} \in Y^*, & h &= Eg = -e_{0,1} - e_{0,2} \in Y, \\ r(a) &= \langle h, a \rangle = -a_{0,1} - a_{0,2}, & La &= -Ea + r(a)h. \end{aligned} \quad (14)$$

Definition 4 (The curve of block sums). For $s > 0$ and $n \geq 1$, write

$$\omega_s(n) = \frac{\max\{1 - sn^{1/4}, 0\}}{4n}, \quad W(s) = \sum_{n \geq 1} \omega_s(n)^2, \quad \sigma = \sum_{n \geq 1} \frac{1}{16n^2} = \frac{\pi^2}{96}. \quad (15)$$

Each ω_s , $s > 0$, has finite support. Set $\omega_0 = (1/(4n))_{n \geq 1} \in \ell^2 \setminus \ell^1$. Define

$$\begin{aligned} J &= (-\infty, -1) \cup (1, \infty), \\ F(t) &= (\text{sgn } t, \omega_{|t|-1}) \in \ell^1(\mathbb{N}_0) \quad (t \in J), \\ D &= \{a \in \ell^1(I) : r(a) \in J, ma = F(r(a))\}. \end{aligned} \quad (16)$$

Since $|t| - 1 > 0$ for $t \in J$, the definition of $F(t)$ uses only ω_s with $s > 0$.

Definition 5 (The two operators on $c_0(I)$). The operators and kernel are

$$\begin{aligned} \text{gra } A &= \{(La, a) : a \in D\}, \\ K &= \{k \in \ell^1(I) : mk = 0, r(k) = 0\}, \\ Px &= \langle x, g \rangle g \quad (x \in Y). \end{aligned} \tag{17}$$

For each $t \in J$, the following is one element of D , not the entire fibre:

$$a^t = -te_{0,1} + (t + \text{sgn } t)e_{0,3} + \sum_{n \geq 1} \omega_{|t|-1}(n)e_{n,1}. \tag{18}$$

We now verify the assumptions of Theorem 1 for the maps in Eqs. (13)–(18). The first three lemmas supply the pairing identity, the exact primal annihilator and the complete parameter fibres.

Lemma 2 (The coordinate pairing identity). *The maps $m : \ell^1(I) \rightarrow \ell^1(\mathbb{N}_0) \subset \ell^2(\mathbb{N}_0)$ and $E, L : \ell^1(I) \rightarrow c_0(I)$ are bounded. For all $a, c \in \ell^1(I)$, Eqs. (19) and (20) below hold.*

Proof. For $a \in \ell^1(I)$, $\|ma\|_1 \leq \|a\|_1$ and $\|Ea\|_\infty \leq 2\|a\|_1$. If a has finite support, then Ea has finite support: only finitely many blocks occur, and within a block no output occurs after the last nonzero input coordinate. Approximation by finite-support vectors and the bound prove $Ea \in c_0(I)$. Thus L also maps $\ell^1(I)$ boundedly into $c_0(I)$.

For $a, c \in \ell^1(I)$, absolute convergence permits exchanging the double sums. In each block the diagonal terms and the two triangular off-diagonal sums give

$$\langle Ea, c \rangle + \langle Ec, a \rangle = 2 \sum_b \left(\sum_j a_{b,j} \right) \left(\sum_j c_{b,j} \right) = 2 \langle ma, mc \rangle_{\ell^2}. \tag{19}$$

The absolute sum of all products is at most $\|a\|_1 \|c\|_1$ before the factor 2, so the identity holds for all $a, c \in \ell^1(I)$. Since $mg = 0$, $h = Eg$, and $r(g) = 0$, Eq. (19) implies $\langle Ea, g \rangle = -r(a)$. Consequently

$$\begin{aligned} \langle La, a \rangle &= r(a)^2 - \|ma\|_2^2, & \langle La, g \rangle &= r(a), \\ \langle La - Lc, a - c \rangle &= (r(a) - r(c))^2 - \|ma - mc\|_2^2. \end{aligned} \tag{20}$$

□

Lemma 3 (The exact primal annihilator). *For K in Eq. (17),*

$$\{z \in c_0(I) : \langle z, k \rangle = 0 \text{ for every } k \in K\} = \mathbb{R}h.$$

Proof. Let $z \in c_0(I)$ annihilate K . In a block $b \geq 1$, every difference $e_{b,j} - e_{b,k}$ is in K . Thus z is constant in that entire block. Since $z \in c_0(I)$, this constant must be zero. In block zero, differences between indices $j, k \geq 3$ belong to K , so the common tail value is also zero. Finally, $g = e_{0,1} - e_{0,2}$ belongs to K , forcing $z_{0,1} = z_{0,2}$. It follows that z is a scalar multiple of h . Conversely, $\langle h, k \rangle = r(k) = 0$ proves that every multiple of h annihilates K . □

Lemma 4 (The tail and the complete parameter fibres). *The map $s \mapsto \omega_s$ from $(0, \infty)$ to $\ell^2(\mathbb{N})$ satisfies Eq. (7) with $\eta = \omega_0$, $S = \sigma < 1/8$ and a Lipschitz constant strictly less than one. For every $t \in J$, the label a^t in Eq. (18) belongs to D , and its entire parameter fibre is $a^t + K$. No $p \in \ell^1(I)$ and $\tau \in [-1, 1]$ satisfy $r(p) = \tau$ and $mp = (\tau, \omega_0)$.*

Proof. For each $s > 0$, $\omega_s(n) = 0$ whenever $n \geq s^{-4}$. Also $0 \leq \omega_s(n) \leq 1/(4n)$. Hence $W(s) \leq \sigma < 1/8$, and dominated convergence in ℓ^2 gives $\omega_s \rightarrow \omega_0$ and $W(s) \rightarrow \sigma$ as s decreases to zero. The comparison vector ω_0 is not in ℓ^1 by divergence of the harmonic series. No endpoint is added to J .

Since $v \mapsto \max\{v, 0\}$ is 1-Lipschitz on $\mathbb{R}$,

$$\|\omega_s - \omega_q\|_2^2 \leq c|s - q|^2, \quad c = \frac{1}{16} \sum_{n \geq 1} n^{-3/2} < \frac{3}{16} < 1. \quad (21)$$

The strict numerical bound follows from $\sum_{n \geq 1} n^{-3/2} < 1 + \int_1^\infty x^{-3/2} dx = 3$. For t, u on the same branch of J , Eq. (21) gives $\|F(t) - F(u)\|_2 \leq |t - u|$. For $t = 1 + s$ and $u = -1 - q$, the squared distance is at most $4 + c(s - q)^2 \leq (2 + s + q)^2$, and the opposite orientation is identical. Thus F is 1-Lipschitz on all of J .

The finitely supported vector a^t in Eq. (18) satisfies $r(a^t) = t$ and $ma^t = F(t)$, proving nonemptiness for every actual t . The complete fibre $\{a : r(a) = t, ma = F(t)\}$ is precisely $a^t + K$. For every $p \in \ell^1(I)$, the vector mp is in $\ell^1(\mathbb{N}_0)$. Since $\omega_0 \notin \ell^1(\mathbb{N})$, the last assertion follows. $\square$

Example 1 (A counterexample on c_0). *Let $I = \mathbb{N}_0 \times \mathbb{N}$ and $Y = c_0(I)$ with its usual supremum norm and full dual $Y^* = \ell^1(I)$. Let $A : Y \rightrightarrows Y^*$ and the bounded positive rank-one map $P : Y \rightarrow Y^*$ be defined by Eq. (17). Then:*

1. *A and P are maximally monotone in $Y \times Y^*$.*
2. *$\text{dom } P = Y$ and $\text{dom } A \neq \emptyset$, so $\text{dom } A \cap \text{int}(\text{dom } P) \neq \emptyset$.*
3. *Every $(x, a) \in \text{gra } A$ satisfies $\|x\|_\infty > 1/2$ and $\langle x, a \rangle \geq -2\sigma\|x\|_\infty$, where $\sigma = \pi^2/96 < 1/8$.*
4. *Every $(x, b) \in \text{gra}(A + P)$ satisfies $\langle x, b \rangle > 1 - \sigma > 7/8$.*
5. *$(0, 0)$ is monotonically related to the entire graph of $A + P$ and does not belong to that graph. Hence $A + P$ is not maximally monotone.*

Proof. Lemmas 2–4 verify every assumption of Theorem 1 with $X = Y$, $M = m$, $H_0 = \ell^2(\mathbb{N})$, $\omega(s) = \omega_s$, $\eta = \omega_0$ and $S = \sigma$. In particular, $g \neq 0$ and $mg = 0$ by Eq. (14); all parameter fibres are present and the exact primal annihilator is $\mathbb{R}h$.

By Eq. (20), the Lipschitz constraint proves monotonicity for every pair of labels in D . The full polar formula (9) and the nonsummability of ω_0 then give $(\text{gra } A)^\mu = \text{gra } A$: no actual $p \in \ell^1(I)$ can satisfy $mp = (\tau, \omega_0)$ for $|\tau| \leq 1$. This proves maximality in $c_0(I) \times \ell^1(I)$.

For every $a \in D$, $t = r(a)$ satisfies $|t| > 1$. By Eq. (20) and $\|g\|_1 = 2$,

$$\|La\|_\infty \geq |t|/2 > 1/2, \quad \langle La, a \rangle = t^2 - 1 - W(|t| - 1) > -\sigma. \quad (22)$$

It follows that $A(0) = \emptyset$ and $\langle La, a \rangle \geq -2\sigma\|La\|_\infty$ for every $a \in D$. In particular $\mathcal{V}(\text{gra } A) \leq 2\sigma < 1/4$. Indeed, for each $a \in D$, Eq. (22) gives $\max\{0, -\langle La, a \rangle\}/\|La\|_\infty \leq \sigma/\|La\|_\infty < 2\sigma$; taking the supremum proves the bound.

The rank-one assertion and maximality of P follow from the direct proof in Theorem 1. Here $\|g\|_1 = 2$, so $\|P\| \leq 4$, and $Pe_{0,1} = g \neq 0$. The actual label $a^2 = -2e_{0,1} + 3e_{0,3}$ gives $La^2 = -6e_{0,1} - 8e_{0,2} - 3e_{0,3}$. Thus $\text{dom } A \cap \text{int}(\text{dom } P) \neq \emptyset$, with $\text{int}(\text{dom } P) = Y$. Every sum point has the form $(La, a + tg)$, $a \in D$ and $t = r(a)$, and

$$\langle -La, -a - tg \rangle = 2t^2 - 1 - W(|t| - 1) > 1 - \sigma > 7/8. \quad (23)$$

This proves that $(0, 0)$ is related to every sum point. The sum is monotone as the sum of two monotone operators; it has no zero primal by Eq. (22). Adding $(0, 0)$ is therefore a proper monotone extension. □

The same graph also has a nonconvex domain closure. This consequence helps identify the relationship between the construction and claims about the domains of arbitrary maximally monotone operators.

Corollary 1. *For the operator A of Example 1, $0 \in \text{conv}(\text{dom } A)$ and $\text{dist}(0, \text{dom } A) \geq 1/2$. In particular, $\overline{\text{dom } A}$ is not convex.*

Proof. At $t = 2$ and $t = -2$, the tails in Eq. (18) vanish, so that $a^{-2} = -a^2$. Since L is linear, both La^2 and $-La^2$ belong to $\text{dom } A$. Their midpoint is zero. Equation (22) gives $\|x\|_\infty > 1/2$ for every $x \in \text{dom } A$, and hence the asserted distance bound. A convex closure containing La^2 and $-La^2$ would contain zero, contradicting this bound. □

4.2 A counterexample on standard ℓ^1

For standard ℓ^1 , let $Z = \ell^1(\mathbb{N})$ and $Z^* = \ell^\infty(\mathbb{N})$, with their usual pairing. Fix once and for all an enumeration (q_n) of all finitely supported rational vectors in the closed unit ball of Y .

Definition 6 (The operators on standard ℓ^1). Define

$$\begin{aligned} Qu &= \sum_{n \geq 1} u_n q_n, & (Q^*a)_n &= \langle q_n, a \rangle, \\ \text{gra } T &= \{(u, Q^*a) : a \in D, Qu = La\}, & & \\ f &= Q^*g, & Bu &= \langle u, f \rangle f. \end{aligned} \quad (24)$$

We prove that every $x \in Y$ has a preimage $u \in Z$ with $Qu = x$ and $\|u\|_1 \leq 2\|x\|_\infty$. No bounded linear right inverse or complemented kernel is assumed. The polar for T and $T + B$ uses all ℓ^∞ labels.

Lemma 5 (The specified quotient onto c_0). *The map $Q : Z = \ell^1(\mathbb{N}) \rightarrow Y = c_0(I)$ of Eq. (24) is a bounded surjection with $\|Q\| = 1$. Its adjoint satisfies $\|Q^*a\|_\infty = \|a\|_1$ for every $a \in \ell^1(I)$. For the fixed enumeration (q_n) , a map $\mathcal{R} : Y \rightarrow Z$ can be specified with*

$$Q\mathcal{R}(x) = x, \quad \|\mathcal{R}(x)\|_1 \leq 2\|x\|_\infty.$$

Proof. Finitely supported rational vectors in the closed unit ball of Y are countable and dense in that ball: truncate a c_0 vector and approximate each retained coordinate by a rational inside $[-1, 1]$. Fix an enumeration of all these vectors as (q_n) . For $u \in Z = \ell^1(\mathbb{N})$, the sum defining Qu converges absolutely in the Banach space Y , and $\|Qu\|_\infty \leq \|u\|_1$. Coordinate unit vectors occur among the q_n , so $\|Q\| = 1$.

For an arbitrary $x \in Y$, set $r_0 = x$. If $r_k \neq 0$, choose the least index n_k for which the actual q_{n_k} satisfies $\|q_{n_k} - r_k / \|r_k\|_\infty\|_\infty < 1/2$ and set

$$c_k = \|r_k\|_\infty, \quad r_{k+1} = r_k - c_k q_{n_k}. \quad (25)$$

If a residual vanishes, stop. Otherwise $\|r_k\|_\infty \leq 2^{-k}\|x\|_\infty$ and $\sum_k c_k \leq 2\|x\|_\infty$. Define $u_n = \sum_{k:n_k=n} c_k$. Repeated indices are combined. Positivity of the c_k and summability give $u \in \ell^1$ with $\|u\|_1 = \sum_k c_k \leq 2\|x\|_\infty$. Telescoping Eq. (25), then absolute convergence under regrouping, proves

$$Qu = x, \quad \|u\|_1 \leq 2\|x\|_\infty. \quad (26)$$

For $x = 0$, use $u = 0$. This proves surjectivity with the asserted lifting bound. The construction does not assert that $x \mapsto u$ is linear or continuous, nor that $\ker Q$ is complemented.

Absolute convergence gives $\langle u, Q^*a \rangle = \langle Qu, a \rangle$ and $\|Q^*a\|_\infty \leq \|a\|_1$. For each finite subset of I , its sign vector for a is a finitely supported rational vector of norm at most one and occurs in the enumeration. Taking these finite sign tests proves

$$\|Q^*a\|_\infty = \|a\|_1 \quad (a \in \ell^1(I)). \quad (27)$$

In particular, Q^* is injective. The least-index choices in Eq. (25) specify

$$\mathcal{R}(x) = \sum_k c_k e_{n_k}, \quad \mathcal{R}(0) = 0.$$

The series converges in ℓ^1 by the bound on $\sum_k c_k$, and Eq. (26) gives both asserted properties. $\square$

For this fixed enumeration,

$$\text{gra } T = \{(\mathcal{R}(La) + v, Q^*a) : a \in D, v \in \ker Q\}.$$

Indeed, $Q\mathcal{R}(La) = La$, and any other preimage differs by an element of $\ker Q$. Thus the entire graph is specified by Eq. (24); its definition does not select a maximal extension. This description asserts neither a bounded linear or continuous right inverse nor decidability of membership in the infinite-dimensional graph.

Example 2 (A counterexample on standard ℓ^1). On $Z = \ell^1(\mathbb{N})$ with its usual norm and full dual $Z^* = \ell^\infty(\mathbb{N})$, let $T : Z \rightrightarrows Z^*$ and $B : Z \rightarrow Z^*$ be defined by Eq. (24). Then:

1. T is maximally monotone in the full pair $Z \times Z^*$.
2. B is nonzero, bounded, positive, rank-one, and maximally monotone, with $\text{dom } B = Z$. In particular, $\text{dom } T \cap \text{int}(\text{dom } B) \neq \emptyset$.
3. $T(0) = \emptyset$, and every $(u, b) \in \text{gra } T$ satisfies $\|u\|_1 > 1/2$ and $\langle u, b \rangle \geq -2\sigma\|u\|_1$. Thus $\mathcal{V}(\text{gra } T) \leq 2\sigma < 1/4$.

4. Every $(u, c) \in \text{gra}(T + B)$ satisfies $\langle u, c \rangle > 1 - \sigma > 7/8$.
5. $(0, 0)$ belongs to the full monotone polar of $\text{gra}(T + B)$ and does not belong to $\text{gra}(T + B)$. Hence $T + B$ is not maximally monotone.

Proof. Lemma 5 gives a surjection Q with the bound $\|\mathcal{R}(x)\|_1 \leq 2\|x\|_\infty$, and Example 1 proves maximality of A . Thus Lemma 1 applies to T . For the specified spaces, we record the full-dual reduction explicitly.

The graph of T is nonempty by Eq. (26) and nonemptiness of D . It is monotone because the bracket of any two graph points (u, Q^*a) and (v, Q^*c) equals $\langle La - Lc, a - c \rangle$, which is nonnegative by Eq. (20) and the Lipschitz bound.

Let (v, v^*) be a polar point with v^* an arbitrary element of $\ell^\infty(\mathbb{N})$. Fix $(u, Q^*a) \in \text{gra } T$. Since $Q(u + \theta k) = Qu$ for $k \in \ker Q$ and $\theta \in \mathbb{R}$, every $(u + \theta k, Q^*a)$ also belongs to $\text{gra } T$. Testing these points forces $\langle k, v^* \rangle = 0$ for every such k . Define a functional on Y by

$$\lambda(x) = \langle u, v^* \rangle \quad \text{when } Qu = x. \quad (28)$$

Two preimages differ by $\ker Q$, proving well-definedness. Linear combinations of preimages prove linearity. For each x , choose the preimage in Eq. (26); then $|\lambda(x)| \leq 2\|v^*\|_\infty\|x\|_\infty$. Thus λ belongs to the continuous dual of $c_0(I)$.

Explicitly put $p_i = \lambda(e_i)$. Finite sign tests give $\sum_{i \in H} |p_i| \leq \|\lambda\|$ for every finite H , whence $p \in \ell^1(I)$. Equality $\lambda(x) = \langle x, p \rangle$ holds first for finite-support x and then for every c_0 vector by density and continuity. Therefore $v^* = Q^*p$: the two functionals agree on every u because Eq. (28) applies to Qu .

Every original $a \in D$ admits a lift u of La by Eq. (26). The full T -polar inequality now reads

$$\langle Qv - La, p - a \rangle \geq 0 \quad \forall a \in D. \quad (29)$$

Example 1 gives $p \in D$ and $Qv = Lp$. Hence $(v, v^*) = (v, Q^*p)$ belongs to $\text{gra } T$. This proves maximality in the full standard pair $\ell^1 \times \ell^\infty$. Membership of v^* in $\text{range}(Q^*)$ was derived from the actual kernel tests.

The radial bound on every graph value. For every $(u, Q^*a) \in \text{gra } T$, with $t = r(a)$,

$$\begin{aligned} \langle u, Q^*a \rangle &= \langle La, a \rangle = t^2 - 1 - W(|t| - 1) > -\sigma, \\ \|u\|_1 &\geq \|Qu\|_\infty = \|La\|_\infty > 1/2. \end{aligned} \quad (30)$$

Thus $T(0) = \emptyset$ and $\langle u, Q^*a \rangle \geq -2\sigma\|u\|_1$ on the entire graph, including every $\ker Q$ translate. Consequently, $\mathcal{V}(\text{gra } T) \leq 2\sigma < 1/4$. No exact equality for this norm-dependent supremum is asserted.

The rank-one factor and the intersection condition. Let $f = Q^*g$. By Eq. (27), $\|f\|_\infty = \|g\|_1 = 2$, so f is nonzero. The map $B : u \mapsto \langle u, f \rangle f$ is nonzero, linear, bounded with norm at most 4, positive, and rank-one. The rank-one proof in Theorem 1, now testing all $v \in \ell^1$ and labels in ℓ^∞ , proves B maximally monotone. Its domain is all of Z , so $\text{int}(\text{dom } B) = Z$.

The vector $q_o = -(3/4)e_{0,1} - e_{0,2} - (3/8)e_{0,3}$ occurs as q_{n_0} . For $u_o = 8e_{n_0}$, $Qu_o = La^2$. Thus $(u_o, Q^*a^2) \in \text{gra } T$, and $\text{dom } T \cap \text{int}(\text{dom } B) \neq \emptyset$ in the stated factor order.

A strict polar witness outside the graph. For every $(u, Q^*a) \in \text{gra } T$, let $t = r(a)$. Then $\langle u, f \rangle = \langle Qu, g \rangle = \langle La, g \rangle = t$. Consequently, for every $(u, Q^*a) \in \text{gra } T$,

$$\langle u, Q^*a + Bu \rangle = 2t^2 - 1 - W(|t| - 1) > 1 - \sigma > 7/8. \quad (31)$$

The sum is monotone by monotonicity of its factors. Equation (31) shows that $(0, 0)$ is related to every point of its graph. Since $T(0) = \emptyset$, $(0, 0) \notin \text{gra}(T + B)$. Adjoining this point gives a proper monotone extension, completing the proof of Example 2 with the standard Banach space, full dual, and ordered intersection condition. No theorem about sums was used to prove either factor maximal. $\square$

The identity $\langle u, Q^*a \rangle = \langle Qu, a \rangle$ transfers the pairing estimates to $T = Q^*AQ$. Thus the origin in $Z \times Z^*$ is monotonically related to $T + B$ and absent from its graph, as the origin in $Y \times Y^*$ is for $A + P$. The two examples use the same construction, with maximality proved in each full dual pair.

5 Conclusions

We gave counterexamples to Rockafellar’s sum conjecture on c_0 and standard ℓ^1 . Our structural theorem computes the entire monotone polar for the stated class of graphs and identifies exactly which equations in the original continuous dual determine maximality. In the coordinate construction, their solutions over the deleted interval would require a nonsummable sequence of block sums. The pairing identity proves monotonicity and the finite radial bound, and shows why adding a full-domain positive rank-one operator makes the sum nonmaximal. The pullback through a bounded surjection gives the second counterexample, with maximality verified in the full $\ell^1 \times \ell^\infty$ pair.

Use of AI

The author supplied earlier constructions and results, involving the sign function and triangular operators, together with obstruction results and a framework for positive rank-one perturbations, and guided their further development to investigate how maximality can fail under addition. Through iterative discussions of these materials and the counterexamples, the author and GPT-5.6 Sol developed a general construction theorem characterizing maximality and identifying when an everywhere-defined rank-one monotone perturbation yields a nonmaximal sum. Under the author’s direction, GPT-5.6 Sol worked out the detailed constructions and proof arguments, including the coupling of triangular operators with a Lipschitz curve whose endpoint limits are not representable in the original continuous dual, and the transfer from c_0 to standard ℓ^1 .

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